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**MATH. EXERCISES.**

# GRADIENT, DIVERGENCE, CURL DEL (NABLA) OPERATOR , LAPLACIAN OPERATOR , CONTINUITY AND NAVIER-STOKES EQUATIONS

## VECTOR PRODUCTS

If $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$ then

$\vec{u} \bullet \vec{v} = u_1v_1 + u_2v_2 + u_3v_3$ (scalar or dot product)

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$ (vector or cross product)

In some books is also considered outer product defined by

$$\vec{u} \otimes \vec{v} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} (v_1 \quad v_2 \quad v_3) = \begin{pmatrix} u_1v_1 & u_1v_2 & u_1v_3 \\ u_2v_1 & u_2v_2 & u_2v_3 \\ u_3v_1 & u_3v_2 & u_3v_3 \end{pmatrix}$$

## GRADIENT, DIVERGENCE, CURL DEL (NABLA) OPERATOR , LAPLACIAN OPERATOR

GRADIENT

Let $\varphi(x, y, z)$ be a scalar field.

The gradient is the vector field defined by

$$grad(\varphi) = (\frac{\partial \varphi}{\partial x}, \frac{\partial \varphi}{\partial y}, \frac{\partial \varphi}{\partial z})$$

DIVERGENCE

Let $\vec{F} = (P(x, y, z), Q(x, y, z), R(x, y, z))$ be a vector field, continuously differentiable with respect to x, y and z.

Then the divergence of $\vec{F}$ is the scalar field defined by

$$div(\vec{F}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

CURL.

The curl of $\vec{F}$ is the vector field defined by

$$curl(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = (\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z})\vec{i} + (\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x})\vec{j} + (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})\vec{k}$$

or $$curl(\vec{F}) = (\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})$$

DEL (NABLA) OPERATOR

The vector differential operator

$$\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z} = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})$$

is called *del* or *nabla* .

Using $\nabla$ we can denote **grad, div** and **curl** as below:

$$grad(\varphi)=\nabla\varphi$$
$$div(\vec{F})=\nabla\bullet\vec{F}$$
$$curl(\vec{F})=\nabla\times\vec{F}$$

Note that $\vec{F}\bullet\nabla$ is not the same as $\nabla\bullet\vec{F}$.

$\vec{F}\bullet\nabla=P\frac{\partial}{\partial x}+Q\frac{\partial}{\partial y}+R\frac{\partial}{\partial z}$.

LAPLACIAN OPERATOR

The Laplacian operator, $\Delta=\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$, is defined for a scalar field $U(x,y,z)$ by

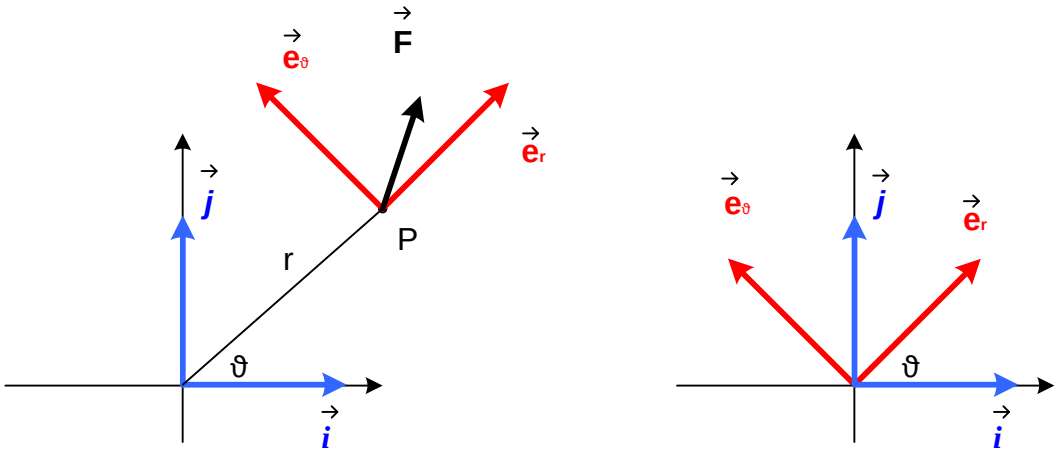
$$\Delta U=\nabla^2U=\frac{\partial^2U}{\partial x^2}+\frac{\partial^2U}{\partial y^2}+\frac{\partial^2U}{\partial z^2},$$

and for a vector field $\vec{F}=(P(x,y,z),Q(x,y,z),R(x,y,z))$ by

$\Delta\vec{F}=\nabla^2\vec{F}=(\Delta P,\Delta Q,\Delta R)$.

## Some formulas for polar and cylindrical coordinates

**Polar coordinates ( 2 dim)**




transformation: $x=r\cos\theta$, $y=r\sin\theta$,

area element: $dA=r\,dr\,d\theta$

standard basis: $\vec{e}_r=\cos\theta\,\vec{i}+\sin\theta\,\vec{j}$, $\vec{e}_\theta=-\sin\theta\,\vec{i}+\cos\theta\,\vec{j}$,

**[Remark 1 :** Note that $\vec{e}_r$, $\vec{e}_\theta$ vary ( depend on $\theta$) when we move from point to point, this is the reason why this basis, in some books, is called " local basis" .]

If $\vec{F}=F_x\vec{i}+F_y\vec{j}$ in Cartesian coord. and $\vec{F}=F_r\vec{e}_r+F_\vartheta\vec{e}_\theta$ the same vector in polar coordinates then

$F_r=F_x\cos\theta+F_y\sin\theta$, $F_\vartheta=-F_x\sin\theta+F_y\cos\theta$.

**[Remark 2:** Vi can derive these formulas by calculating the components of $\vec{F}$ in the directions of $\vec{e}_r$ and $\vec{e}_\theta$. Thus

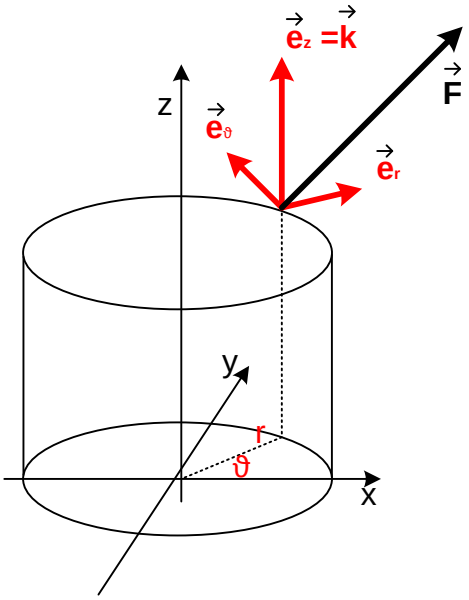
$F_r = \vec{F} \bullet \vec{e}_r = (F_x\vec{i} + F_y\vec{j}) \bullet (\cos\theta\,\vec{i} + \sin\theta\,\vec{j}) = F_x\cos\theta + F_y\sin\theta$, similarly

$F_\vartheta = \vec{F} \bullet \vec{e}_\vartheta = (F_x\vec{i} + F_y\vec{j}) \bullet (-\sin\theta\,\vec{i} + \cos\theta\,\vec{j}) = -F_x\sin\theta + F_y\cos\theta$. ]

## Cylindrical coordinates $(r,\theta,z)$:

transformation: $x = r\cos\theta$, $y = r\sin\theta$, z=z

volume element: $dV = r\,dr\,d\theta\,dz$




standard basis: $\vec{e}_r = \cos\theta\,\vec{i} + \sin\theta\,\vec{j}$, $\vec{e}_\theta = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}$, $\vec{e}_z = \vec{k}$

If $\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$ in Cartesian coord. and $\vec{F} = F_r\vec{e}_r + F_\vartheta\vec{e}_\theta + F_z\vec{k}$ the same vector in cylindrical coordinates then we have following vector components relationship:

$F_r = F_x\cos\theta + F_y\sin\theta$, $F_\vartheta = -F_x\sin\theta + F_y\cos\theta$, $F_z = F_z$

**[Remark 3:** For example, we can get $F_r = F_x\cos\theta + F_y\sin\theta$ in the following way:

$F_r = \vec{F} \bullet \vec{e}_r = (F_x\vec{i} + F_y\vec{j} + F_z\vec{k}) \bullet (\cos\theta\,\vec{i} + \sin\theta\,\vec{j}) = F_x\cos\theta + F_y\sin\theta$ ]

scalar field: $f(r,\theta,z)$

gradient: $grad(f) = \nabla f = \dfrac{\partial f}{\partial r}\vec{e}_r + \dfrac{1}{r}\dfrac{\partial f}{\partial\theta}\vec{e}_\theta + \dfrac{\partial f}{\partial z}\vec{e}_z$

laplacian: $\nabla^2 f = \Delta f = \dfrac{1}{r}\dfrac{\partial}{\partial r}\left(r\dfrac{\partial f}{\partial r}\right) + \dfrac{1}{r^2}\dfrac{\partial^2 f}{\partial\theta^2} + \dfrac{\partial^2 f}{\partial z^2}$

vector field: $\vec{F} = (F_r, F_\theta, F_z)$

divergence: $div(\vec{F}) = \nabla \circ \vec{F} = \dfrac{1}{r}\dfrac{\partial(r\cdot F_r)}{\partial r} + \dfrac{1}{r}\dfrac{\partial(F_\theta)}{\partial\theta} + \dfrac{\partial F_z}{\partial z}$

curl: $curl(\vec{F}) = \nabla \times \vec{F} = \dfrac{1}{r}\begin{vmatrix} \vec{e}_r & r\cdot\vec{e}_\theta & \vec{e}_z \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial\theta} & \dfrac{\partial}{\partial z} \\ F_r & r\cdot F_\theta & F_z \end{vmatrix} =$

$$\left(\frac{1}{r}\frac{\partial F_z}{\partial\theta} - \frac{\partial F_\theta}{\partial z}\right)\vec{e}_r + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}\right)\vec{e}_\theta + \frac{1}{r}\left(\frac{\partial(rF_\theta)}{\partial r} - \frac{\partial F_r}{\partial\theta}\right)\vec{e}_k$$

**EXERCISES**

**1.** Find

a) $div(\vec{F})$ , b) $grad(div(\vec{F}))$ and c) $curl(\vec{F})$

if $\vec{F} = (y + x^2, z, x^2)$

**2.** Prove the following identities

**a )** $div(curl(\vec{F}) = 0$ (or $\nabla \bullet (\nabla \times \vec{F}) = 0$ **)**

**b)** $curl(div(\varphi)) = \vec{0}$ (or $\nabla \times (\nabla \varphi) = \vec{0}$ **)**

**3.** Which one of the following functions

a) $f_1(x, y, z) = 3x^2 + 2y^2 + z^2$

b) $f_2(x, y, z) = x + \ln(y^2 + z^2)$

c) $f_3(x, y, z) = x^3 + \exp(y + z)$

d) $f_4(x, y, z) = 5x^2 - 5y^2 + x + y + 4z$

satisfies the Laplace equation $\Delta f$ =0 ?

**4.** Find $\Delta f + \nabla \bullet (\nabla \times (\nabla f)))$

if $f(x, y, z) = x^3 + y^2 + z$.

**5.** Write the general transport equation

$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot \nabla\varphi) + S_\phi$$

without using operators div, $\nabla$, $\Delta$, curl or grad.

Here $\vec{U} = (u, v, w)$. Functions $\rho$, $\varphi$, $\Gamma$, $S$ , u, v, w are real functions of *t*, *x*, *y* and *z*.

**6.** Which one, if any, of the following functions

a) $\varphi_1(x, y, z) = x^4 + y^2 + z$

b) $\varphi_2(x, y, z) = x^2 + y^2 + z$

c) $\varphi_3(x, y, z) = x^2 + y^2 + z^2$

satisfies the equation

$\nabla \bullet (\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + S$ ?

Here $\Gamma = 5$ , $\vec{U} = (1, 2, 3)$ and $S = 2x + 4y - 23$.

**7.** Find which one (if any) of the following functions

a) $\varphi_1(x, y, z) = x^2 + y^2 + z^2$

b) $\varphi_2(x, y, z) = x^2 + y^2 + 5z$

c) $\varphi_3(x, y, z) = 5x^2 + y^2 + z^2$

satisfies the equation

$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi\vec{U}) = div(\Gamma grad\varphi) + S$$

where ρ=3, $\Gamma = 2$ , $\vec{U} = (2, 3, 4)$ and $S = 12x + 18y + 52$.

8. (exam 1, 2008)

**A)** Write the general transport equation

$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + S \qquad (\text{eq } 1)$$

without using operators div, $\nabla$, $\Delta$, curl or grad.

Here $\vec{U} = (u, v, w)$. Functions $\rho$, $\varphi$, $\Gamma$, $S$ , u, v, w are real functions of *t, x, y* and *z*.

**B)** Let $\rho = 2$, $\Gamma = 3$ , $\vec{U} = (1, 2, 4)$.

Find S in the equation (eq 1) if we now that the function

$\varphi(x, y, z) = x + y^2 + z^3$

satisfies the equation.

**9. (Q6, exam 2, 2008)**

Consider the following equation

$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + \nabla \bullet (\nabla \times \vec{U}) + 6y - 2xyz - 4 \qquad (\text{eq } 1)$$

Let $\rho = 1$, $\Gamma$ = constant , $\vec{U} = (2, 3, -xy)$.

Find the constant $\Gamma$ in the equation (eq 1) if we now that the function

$\varphi(x, y, z) = 2t + x + y^2 + z^2$

satisfies the equation.

**10.** If possible, find $f(x, y)$ for the given partial derivatives $\frac{\partial}{\partial x} f(x, y)$ and $\frac{\partial}{\partial y} f(x, y)$.

a) $\frac{\partial}{\partial x} f(x, y) = 2xy$ and $\frac{\partial}{\partial y} f(x, y) = x^2 + 2y$.

b) $\frac{\partial}{\partial x} f(x, y) = 2x + y$ and $\frac{\partial}{\partial y} f(x, y) = x$.

c) $\frac{\partial}{\partial x} f(x, y) = ye^{xy}$ and $\frac{\partial}{\partial y} f(x, y) = xe^{xy}$.

d) $\frac{\partial}{\partial x} f(x, y) = 2x + y$ and $\frac{\partial}{\partial y} f(x, y) = 5x$.

**( Hint: Necessary condition: If** $f(x, y)$ has continues derivatives then the mixed derivatives of $f(x, y)$ should be equal.

Thus

$$\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x} f(x, y\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y} f(x, y\right) \qquad (*)$$

is the necessary condition for the existence of a function $f(x, y)$ that has the given derivatives.

**11.** Determine the value of *a* for which the system of partial differential equations

$\frac{\partial}{\partial x} f(x,y) = axy + y$ and $\frac{\partial}{\partial y} f(x,y) = x^2 + x$.
has solutions. Then find $f(x,y)$ corresponding to this value of *a*.

**12.** If possible, find $f(x,y,z)$ for the given partial derivatives $\frac{\partial}{\partial x} f(x,y,z)$, $\frac{\partial}{\partial y} f(x,y,z)$ and $\frac{\partial}{\partial z} f(x,y,z)$.

a) $\frac{\partial}{\partial x} f(x,y,z) = yz$, $\frac{\partial}{\partial y} f(x,y,z) = xz + z$ and $\frac{\partial}{\partial z} f(x,y,z) = xy + y + 3z^2$

b) $\frac{\partial}{\partial x} f(x,y,z) = yz$, $\frac{\partial}{\partial y} f(x,y,z) = xz + z$ and $\frac{\partial}{\partial z} f(x,y,z) = xy + y + 3z^2$

c) $\frac{\partial}{\partial x} f(x,y,z) = yze^{xyz}$, $\frac{\partial}{\partial y} f(x,y,z) = xze^{xyz}$ and $\frac{\partial}{\partial z} f(x,y,z) = xye^{xyz}$

d) $\frac{\partial}{\partial x} f(x,y,z) = yz$, $\frac{\partial}{\partial y} f(x,y,z) = xz + z$ and $\frac{\partial}{\partial z} f(x,y,z) = xy + y + xz$

**( Hint: Necessary condition: If** $f(x,y,z)$ has continuous derivatives then the mixed derivatives of $f(x,y)$ should be equal.
Thus

$Con1: \quad \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x} f\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y} f\right)$

$Con2: \quad \frac{\partial}{\partial z}\left(\frac{\partial}{\partial x} f\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial z} f\right)$

$Con3: \quad \frac{\partial}{\partial z}\left(\frac{\partial}{\partial y} f\right) = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial z} f\right)$

are the necessary condition for the existence of a function $f(x,y)$ that has the given derivatives.

**13.** Determine the values of *a* and *b* for which the system of partial differential equations

$\frac{\partial}{\partial x} f(x,y,z) = ax^2yz + 2x$, $\frac{\partial}{\partial y} f(x,y,z) = x^3z + 1$ and $\frac{\partial}{\partial z} f(x,y,z) = bx^3y + 2z$
has solutions. Then find $f(x,y,z)$ corresponding to these values of *a* and *b*.

**14.**
We consider an incompressible ( density $\rho$ =const), steady state ( variables do not depend on time), isothermal Newtonian flow with a given velocity field
$\vec{V} = (u(x,y,z),\ v(x,y,z),\ w(x,y,z))$.
Use the following equations ( continuity and Navier Stokes equations) to find en expression for pressure P(*x*,*y*,*z*) as a function of *x*,*y* and *z*, where
$\rho$ =constant, $\mu$ =constant , $\vec{g} = (0,0,-g)$ i.e. $g_x = g_y = 0$ and
$g_z = -g$ (where $g \approx 9.81 m/s^2$)

**Incompressible continuity equation:**

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \qquad \textbf{eq1.}$$

**Navier Stokes equations:**

*x component:*

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=-\frac{\partial P}{\partial x}+\rho g_x+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right) \qquad \textbf{eq2.}$$

*y component:*

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=-\frac{\partial P}{\partial y}+\rho g_y+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right) \qquad \textbf{eq3.}$$

*z component:*

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=-\frac{\partial P}{\partial z}+\rho g_z+\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right) \qquad \textbf{eq4.}$$

**a)** $\vec{V}=(2x+3y,\ 4x-2y,\ 0)$

**b)** $\vec{V}=(3x+4y,\ 2x-3y,\ -2)$

**c)** $\vec{V}=(1+2y,\ -4x,\ 2)$

**15. (exam 1, 2009)**

**A)** Consider the following equation

$$\frac{\partial(\rho\varphi)}{\partial t}+\nabla\bullet(\rho\varphi\vec{U})=\nabla\bullet(\Gamma\cdot(\nabla\varphi))+\nabla\bullet(\nabla\times\vec{U})+16x+16y+4xz+8yz-24 \qquad (\text{eq 1})$$

Let $\rho=2$, $\Gamma=$ constant, $\vec{U}=(4,\ 4,\ x+2y)$.

Find the constant $\Gamma$ in the equation (eq 1) if we now that the function

$\varphi(x,y,z)=1+3t+x^2+y^2+z^2$

satisfies the equation.

**B)** We consider an incompressible ( density $\rho$ =const), steady state ( variables do not depend on time), isothermal Newtonian flow with a given velocity field

$\vec{V}=(u(x,y,z),\ v(x,y,z),\ w\ (x,y,z))$.

Use the following equations ( continuity and Navier Stokes equations) to find en expression for pressure P(*x,y,z*) as a function of *x,y* and *z*, where

$\rho$ =constant, $\mu$ =constant ,

$\vec{g}=(0,0,-g)$ i.e. $g_x=g_y=0$ and $g_z=-g$ (where $g\approx 9.81 m/s^2$) and

$\vec{V}=(6+4x,\ 4-2y,\ 2-2z)$ **.**

**16. (exam 2, 2009)**

We consider an incompressible ( density $\rho$ =const), steady state ( variables do not depend on time), isothermal Newtonian flow with a given velocity field

$\vec{V}=(u(x,y,z),\ v(x,y,z),\ w\ (x,y,z))$.

Use the following equations ( continuity and Navier Stokes equations) to find first

i) parameter *a*

and then

ii) en expression for pressure P($x,y,z$) as a function of $x,y$ and $z$, where
$\rho$ =constant, $\mu$ =constant ,
$\vec{g} = (0,0,-g)$ i.e. $g_x = g_y$ =0 and $g_z = -g$ (where $g \approx 9.81 m/s^2$) and
$\vec{V} = (2+3x, 5-y, 1-az)$ **.**

**17.** Consider steady, incompressible, isothermal, laminar stationary Newtonian flow in a long round pipe in the z-direction, with constant circular cross-section of radius R=2 m. Use the continuity and the Navier-Stokes equations in cylindrical coordinates to find the velocity field **V**=($u_r$, $u_\theta$, $u_z$) and the pressure field P (r,θ,z) if the fluid flow satisfies the following conditions:

**c0.** All partial derivatives with respect to time t are 0 **( Steady flow)**

**c1.** μ=0.001 kg/(m·s) and $\rho$ =1000 kg/m$^3$

**c2.** A Constant pressure gradient $\partial P/\partial z = -1/250$ Pa/m is applied in the horizontal axis ( z-axis in our notation):
$\partial P/\partial z = -1/250$,

**c3.** The flow is parallel to the z axis, that is $u_r$ =0 and $u_\theta$ =0.

**c4.** We assume that the flow is axisymmetric . The velocity does not depend on θ, that is $\frac{\partial u_z}{\partial \theta} = 0$

**c5.** Boundary cond. 1 ( No-slip boundary condition, $\mathbf{V}_{fluid}=\mathbf{V}_{wall}$ ): If r=2 then $u_z$= 0

**c6.** Boundary condition 2: $u_z$ has maximum at r=0 that is $\left.\frac{\partial u_z}{\partial r}\right|_{r=0} = 0$

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*The continuity and the Navier-Stokes equations for an incompressible , isothermal Newtonian flow (density $\rho$ =const, viscosity $\mu$ =const), with a velocity field* $\vec{V} = (u_r, u_\theta, u_z)$ ***in Cylindrical coordinates*** $(r,\theta,z)$ *:*

**Incompressible continuity equation**

$$\frac{1}{r}\frac{\partial (r u_r)}{\partial r} + \frac{1}{r}\frac{\partial (u_\theta)}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0 \qquad \textbf{eq a)}$$

**Navier-Stokes equations in Cylindrical coordinates:**

*r-component:*

$$\rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta^2}{r} + u_z\frac{\partial u_r}{\partial z}\right)$$

$$= -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right) - \frac{u_r}{r^2} + \frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial^2 u_r}{\partial z^2}\right] \qquad \textbf{eq b)}$$

*θ -component:*

$$\rho\left(\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r u_\theta}{r}+u_z\frac{\partial u_\theta}{\partial z}\right)$$
$$=-\frac{1}{r}\frac{\partial P}{\partial \theta}+\rho g_\theta+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\theta}{\partial r}\right)-\frac{u_\theta}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2}+\frac{2}{r^2}\frac{\partial u_r}{\partial \theta}+\frac{\partial^2 u_\theta}{\partial z^2}\right] \qquad \textbf{eq c)}$$

*z-component:*

$$\rho\left(\frac{\partial u_z}{\partial t}+u_r\frac{\partial u_z}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta}+u_z\frac{\partial u_z}{\partial z}\right)$$
$$=-\frac{\partial P}{\partial z}+\rho g_z+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2}+\frac{\partial^2 u_z}{\partial z^2}\right] \qquad \textbf{eq d)}$$

**18. (Exam 1 March 2012, question A , 4points.)**

We consider an incompressible ( density $\rho$ =const), steady state ( variables do not depend on time), isothermal Newtonian flow with a given velocity field

$\vec{V} = (u(x,y,z),\ v(x,y,z),\ w\ (x,y,z)) = (3x + cy,\ 4 + 2x - y + bz,\ 3 - az)$

Use the following equations ( continuity and Navier Stokes equations) , where

$\rho$ =constant, $\mu$ =constant , $\vec{g} = (0,0,-g)$ i.e. $g_x = g_y$ =0 and $g_z = -g$ ( $g \approx 9.81 m/s^2$)

to find:

i) parameters a, b and c

ii) en expression for pressure P(*x,y,z*) as a function of *x,y* and *z*.

**The GRADIENT VECTOR with change of variables and basis.**

The gradient vector for the function *f*(*x,y,z*) is defined as

$$grad(f) = \left(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y},\frac{\partial f}{\partial z}\right) = \frac{\partial f}{\partial x}\vec{i}+\frac{\partial f}{\partial y}\vec{j}+\frac{\partial f}{\partial z}\vec{k} \quad (*).$$

If we change variables x, v, z to u, v , w and replace basis vectors $\vec{i}, \vec{j}, \vec{k}$ with new ( linearly independent) vectors $\vec{e}_1, \vec{e}_2, \vec{e}_3$ then we can express the same gradient vector $grad(f)$ in terms of variables u, v , w and vectors $\vec{e}_1, \vec{e}_2, \vec{e}_3$.

We simple calculate the derivatives $\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$ in new variables and express $\vec{i}, \vec{j}, \vec{k}$ as a linear combinations of $\vec{e}_1, \vec{e}_2, \vec{e}_3$. Then we substitute those values into (*).

(See the following example.)

**19.** We consider a scalar field $f(r,\theta,z)$ given in cylindrical coordinates, where

$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$, and basis vectors are $\vec{e}_1, \vec{e}_2, \vec{e}_3$.

Find the expression for the gradient, $grad(f(r,\theta,z))$, in cylindrical coordinates, that is in terms of $r,\theta,z,\vec{e}_1,\ \vec{e}_2,\ \vec{e}_2,\frac{\partial f}{\partial r},\frac{\partial f}{\partial \theta}$ and $\frac{\partial f}{\partial z}$ if

**a)** $\vec{e}_1=\vec{i},\quad \vec{e}_2=\vec{j},\quad \vec{e}_3=\vec{k}$ ( we keep the same basis $\vec{i},\vec{j},\vec{k}$ ) .

**b)** $\vec{e}_1=\vec{i}\quad \vec{e}_2=2\vec{j},\quad \vec{e}_3=2\vec{j}+\vec{k}$

**c)** $\vec{e}_1=\vec{i}\cos\theta,\quad \vec{e}_2=\vec{j}\sin\theta,\ \vec{e}_3=\vec{k}$ **(exam 1, 2012; Q5 B (2 points))**

**d)** $\vec{e}_1=\cos\theta\,\vec{i}+\sin\theta\,\vec{j},\quad \vec{e}_2=-\sin\theta\,\vec{i}+\cos\theta\,\vec{j},\quad \vec{e}_3=\vec{k}$ (this is often used as a local basis for cylindrical coordinates)

**20. (exam 2016; Q6 A (2 points))** Derive the Cauchy momentum equation

$$\rho\frac{D\vec{V}}{Dt}=\nabla\cdot\boldsymbol{\sigma}+\rho\vec{g}\ .$$

**21.** Find the flux of the vector field $\vec{F}=x\vec{i}+y\,\vec{j}+(x+y)\vec{k}$ upward through the surface $z=1+3x+2y$, $0\le x\le 1$, $0\le y\le 2$.

**22.** Use the Divergence Theorem to find the flux of the vector field $\vec{F}=12x\vec{i}+2yz\,\vec{j}-z^2\vec{k}$ out of the sphere S with equation $x^2+y^2+z^2=9$

ANSWERS AND SOLUTIONS:

**1. Solution:**

a) Since $div(\vec{F})=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}$

we have $\vec{F}=(y+x^2,z,x^2)\ \Rightarrow div(\vec{F})=2x+0+0=2x$ .

**Answer** a) $div(\vec{F})=2x$

b) Since $grad(\varphi)=(\frac{\partial\varphi}{\partial x},\frac{\partial\varphi}{\partial y},\frac{\partial\varphi}{\partial z})$ we have ( for $\varphi=div(\vec{F})$ )

$grad(div(\vec{F}))=(2,0,0)$

**Answer** b) $grad(div(\vec{F}))=(2,0,0)$

c) $curl(\vec{F})\overset{def}{=}\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ P & Q & R\end{vmatrix}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ y+x^2 & z & x^2\end{vmatrix}$

$=-1\vec{i}-2x\vec{j}-\vec{k}=(-1,-2x,-1)$

**Answer** c) $curl(\vec{F}) = (-1,-2x,-1)$

**2.** Hint. Use the definitions of div and curl.

**3.**

$$\Delta f = 0 \quad \Leftrightarrow \quad \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

**Answer:** The functions

$f_2(x, y, z) = x + \ln(y^2 + z^2)$ and

$f_4(x, y, z) = 5x^2 - 5y^2 + x + y + 4z$

satisfies the Laplace equation.

**4. Answer:**

$$\Delta f + \nabla \bullet (\nabla \times (\nabla f))) = \Delta f + div(curl(gradf))) = 6x + 2$$

**5. Solution:**

$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi\vec{U}) = div(\Gamma grad\varphi) + S_\phi \Rightarrow$$

$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi u,\ \rho\varphi v,\ \rho\varphi w) = div(\Gamma\frac{\partial\varphi}{\partial x}, \Gamma\frac{\partial\varphi}{\partial y}, \Gamma\frac{\partial\varphi}{\partial z}) + S_\phi \Rightarrow$$

$$\frac{\partial(\rho\varphi)}{\partial t} + \frac{\partial(\rho\varphi u)}{\partial x} + \frac{\partial(\rho\varphi v)}{\partial y} + \frac{\partial(\rho\varphi w)}{\partial z}) = \frac{\partial}{\partial x}\left(\Gamma\frac{\partial\varphi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\Gamma\frac{\partial\varphi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\Gamma\frac{\partial\varphi}{\partial z}\right) + S_\phi$$

**6.**

Which one (if any) of the following functions

a) $\varphi_1(x, y, z) = x^4 + y^2 + z$

b) $\varphi_2(x, y, z) = x^2 + y^2 + z$

c) $\varphi_3(x, y, z) = x^2 + y^2 + z^2$

satisfies the equation

$\nabla \bullet (\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + S$ ?

Here $\Gamma = 5$ , $\vec{U} = (1, 2, 3)$ and $S = 2x + 4y - 23$.

**Solution :**

The equation

$\nabla \bullet (\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + S$

can be written as

$$div(\varphi\vec{U}) = div(\Gamma grad\varphi) + S \Rightarrow$$

$$div(\varphi, 2\varphi, 3\varphi) = div(\frac{5\partial\varphi}{\partial x}, \frac{5\partial\varphi}{\partial y}, \frac{5\partial\varphi}{\partial z}) + 2x + 4y - 23 \Rightarrow$$

$$\frac{\partial\varphi}{\partial x} + \frac{2\partial\varphi}{\partial y} + \frac{3\partial\varphi}{\partial z} = \frac{5\partial^2\varphi}{\partial x^2} + \frac{5\partial^2\varphi}{\partial y^2} + \frac{5\partial^2\varphi}{\partial z^2} + 2x + 4y - 23 \qquad (\text{eq}\,1.)$$

a) Let $\varphi = \varphi_1(x, y, z) = x^4 + y^2 + z$

Vi calculate the derivatives of $\varphi_1$ and substitute in the left hand side (LHS) and right hand side of the equation (eq1).

LHS: $\frac{\partial\varphi}{\partial x} + \frac{2\partial\varphi}{\partial y} + \frac{3\partial\varphi}{\partial z} = 4x^3 + 4y + 3$

RHS= $\frac{5\partial^2\varphi}{\partial x^2} + \frac{5\partial^2\varphi}{\partial y^2} + \frac{5\partial^2\varphi}{\partial z^2} + 2x + 4y - 23 = 60x^2 + 2x + 4y - 13$

Whence $LHS \neq RHS$

Thus the function $\varphi_1(x, y, z) = x^4 + y^2 + z$ is not a solution to the equation

b) $\varphi = \varphi_2(x, y, z) = x^2 + y^2 + z$

LHS= $3 + 2x + 4y$ , RHS= $-3 + 2x + 4y$

Whence $LHS \neq RHS$, and the function $\varphi_2(x, y, z) = x^2 + y^2 + z$ is not a solution to the equation

c) Let $\varphi = \varphi_3(x, y, z) = x^2 + y^2 + z^2$

Then

LHS= $2x + 4y + 6z$ , RHS= $7 + 2x + 4y$

Thus $LHS \neq RHS$, and the function $\varphi_3(x, y, z) = x^2 + y^2 + z^2$ is not a solution to the equation.

**Answer:** None of the functions satisfies the equation

**7. Answer:** Function $\varphi_2(x, y, z) = x^2 + y^2 + 5z$ satisfies the equation.

**8. (exam 1, 98)**

**A)** Write the general transport equation

$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + S \qquad (\text{eq } 1)$$

without using operators div, $\nabla$, $\Delta$, curl or grad.

Here $\vec{U} = (u, v, w)$. Functions $\rho$, $\varphi$, $\Gamma$, $S$ , u, v, w are real functions of *t*, *x*, *y* and *z*.

**B)** Let $\rho = 2$, $\Gamma = 3$, $\vec{U} = (1, 2, 4)$.
Find S in the equation (eq 1) if we now that the function
$\varphi(x, y, z) = x + y^2 + z^3$
satisfies the equation.

**Solution:**
**A)**
$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + S \Rightarrow$$
$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi\vec{U}) = div(\Gamma grad\varphi) + S \Rightarrow$$
$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi u, \rho\varphi v, \rho\varphi w) = div(\Gamma\frac{\partial\varphi}{\partial x}, \Gamma\frac{\partial\varphi}{\partial y}, \Gamma\frac{\partial\varphi}{\partial z}) + S \Rightarrow$$
$$\frac{\partial(\rho\varphi)}{\partial t} + \frac{\partial(\rho\varphi u)}{\partial x} + \frac{\partial(\rho\varphi v)}{\partial y} + \frac{\partial(\rho\varphi w)}{\partial z}) = \frac{\partial}{\partial x}\left(\Gamma\frac{\partial\varphi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\Gamma\frac{\partial\varphi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\Gamma\frac{\partial\varphi}{\partial z}\right) + S \qquad \text{(eq2)}$$
**B)**
We substitute $\rho = 2$, $\Gamma = 3$, $\vec{U} = (1, 2, 4)$ and $\varphi(x, y, z) = x + y^2 + z^3$
in the equation (eq2) and get

$$0 + \frac{\partial(2\varphi)}{\partial x} + \frac{\partial(4\varphi)}{\partial y} + \frac{\partial(8\phi)}{\partial z}) = \frac{\partial}{\partial x}\left(3\frac{\partial\varphi}{\partial x}\right) + \frac{\partial}{\partial y}\left(3\frac{\partial\varphi}{\partial y}\right) + \frac{\partial}{\partial z}\left(3\frac{\partial\varphi}{\partial z}\right) + S$$
$0 + 2 + 8y + 24z^2 = 0 + 6 + 18z + S$.
Consequently
$S = -4 + 8y - 18z + 24z^2$

**9. (Q6, exam 2, 2008)**
Consider the following equation
$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + \nabla \bullet (\nabla \times \vec{U}) + 6y - 2xyz - 4 \qquad \text{( eq 1)}$$
Let $\rho = 1$, $\Gamma$ = constant, $\vec{U} = (2, 3, -xy)$.
Find the constant $\Gamma$ in the equation (eq 1) if we now that the function
$\varphi(x, y, z) = 2t + x + y^2 + z^2$
satisfies the equation.

**Solution:**
$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + \nabla \bullet (\nabla \times \vec{U}) + 6y - 2xyz - 4 \Rightarrow$$
$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi\vec{U}) = div(\Gamma grad\varphi) + div(curl(\vec{U})) + 6y - 2xyz - 4 \Rightarrow$$
(since $curl(\vec{U}) = (-x, y, 0)$ we have $div(curl(\vec{U})) = -1 + 1 = 0$)

$$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi u,\ \rho\varphi v,\ \rho\varphi w) = div(\Gamma\frac{\partial\varphi}{\partial x}, \Gamma\frac{\partial\varphi}{\partial y}, \Gamma\frac{\partial\varphi}{\partial z}) + 0 + 6y - 2xyz - 4 \Rightarrow$$

$$\frac{\partial(\rho\varphi)}{\partial t} + \frac{\partial(\rho\varphi u)}{\partial x} + \frac{\partial(\rho\varphi v)}{\partial y} + \frac{\partial(\rho\varphi w)}{\partial z}) = \frac{\partial}{\partial x}\left(\Gamma\frac{\partial\varphi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\Gamma\frac{\partial\varphi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\Gamma\frac{\partial\varphi}{\partial z}\right) + 0 + 6y - 2xyz - 4 \qquad \text{(eq2)}$$

We substitute $\rho = 1$, , $\vec{U} = (2, 3, -xy)$ and $\varphi(x, y, z) = 2t + x + y^2 + z^2$
in the equation (eq2) and get

$$\frac{\partial(1\varphi)}{\partial t} + \frac{\partial(2\varphi)}{\partial x} + \frac{\partial(3\varphi)}{\partial y} + \frac{\partial(-xy\phi)}{\partial z}) = \frac{\partial}{\partial x}\left(\Gamma\frac{\partial\varphi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\Gamma\frac{\partial\varphi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\Gamma\frac{\partial\varphi}{\partial z}\right) + 6y - 2xyz - 4$$

( Note that $\Gamma$ is a constant)

$2 + 2 + 6y - 2xyz = 0 + 2\Gamma + 2\Gamma + 6y - 2xyz - 4 \Rightarrow$

$8 = 4\Gamma \Rightarrow$

$\Gamma = 2$

Answer: $\Gamma = 2$

**10.** If possible, find $f(x, y)$ for the given partial derivatives $\frac{\partial}{\partial x} f(x, y)$ and $\frac{\partial}{\partial y} f(x, y)$.

a) $\frac{\partial}{\partial x} f(x, y) = 2xy$ and $\frac{\partial}{\partial y} f(x, y) = x^2 + 2y$.

b) $\frac{\partial}{\partial x} f(x, y) = 2x + y$ and $\frac{\partial}{\partial y} f(x, y) = x$.

c) $\frac{\partial}{\partial x} f(x, y) = ye^{xy}$ and $\frac{\partial}{\partial y} f(x, y) = xe^{xy}$.

d) $\frac{\partial}{\partial x} f(x, y) = 2x + y$ and $\frac{\partial}{\partial y} f(x, y) = 5x$.

**( Hint: Necessary condition: If** $f(x, y)$ has continuous derivatives then the mixed derivatives of $f(x, y)$ should be equal, i.e.

$$\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x} f(x, y)\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y} f(x, y)\right) \qquad (*)$$

is the necessary condition for the existence of a function $f(x, y)$ that has the given derivatives.

**Answer:**

a) $f(x, y) = x^2 y + y^2 + C$ b) $f(x, y) = x^2 + xy + C$ c) $f(x, y) = e^{xy} + C$

d) No solution since the condition (*) is not fulfilled,

$$1 = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x} f(x, y)\right) \neq \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y} f(x, y)\right) = 5.$$

**Solution a)**

Since $\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x} f(x, y)\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y} f(x, y)\right)$ =2x and the derivatives are continuous the condition (*) is fulfilled and we can find $f(x, y)$ for the given derivatives.

In order to find $f(x, y)$ we integrate with respect to x the first of the equations

$$\frac{\partial}{\partial x} f(x, y) = 2xy \qquad \text{(eq1)}$$

$\frac{\partial}{\partial y} f(x,y) = x^2 + 2y$ (eq2).

and get

$f(x,y) = \int 2xydx = x^2 y + C_1(y)$

Thus

$f(x,y) = x^2 y + C_1(y)$ ( i )

We have integrated with respect to *x*, therefore the constant still depend on *y*.

Now, to find $C_1(y)$ we differentiate and substitute (i) in (eq2) and get:

$\frac{\partial}{\partial y}\left(x^2 y + C_1(y)\right) = x^2 + 2y \Rightarrow x^2 + \frac{\partial}{\partial y}\left(C_1(y)\right) = x^2 + 2y \Rightarrow$

$\frac{\partial}{\partial y}\left(C_1(y)\right) = 2y \Rightarrow C_1(y) = y^2 + C$ .

Finally, substituting $C_1(y) = y^2 + C$ in (i) we have

$f(x,y) = x^2 y + y^2 + C$ (where C is a constant).

**11.**

**Answer:** From $\frac{\partial}{\partial y}\frac{\partial}{\partial x} f(x,y) = \frac{\partial}{\partial x}\frac{\partial}{\partial y} f(x,y) \Rightarrow ay + 1 = 2x + 1 \Rightarrow a = 2$

Then for $a = 2$ we have $f(x,y) = x^2 y + xy + C$

**12.**

**Answer:**

a) $f = xyz + yz + z^3 + C$ b) $f = xy + yz + C$ c) $f = e^{xyz} + C$

d) No solution since the condition *Con*2 is not fulfilled,

$y = \frac{\partial}{\partial z}\left(\frac{\partial}{\partial x} f\right) \neq \frac{\partial}{\partial x}\left(\frac{\partial}{\partial z} f\right) = y + z$

**Solution a)**

a) $\frac{\partial}{\partial x} f(x,y,z) = yz$, $\frac{\partial}{\partial y} f(x,y,z) = xz + z$ and $\frac{\partial}{\partial z} f(x,y,z) = xy + y + 3z^2$

Since the conditions Con1,2,3 are fulfilled and we can find $f(x,y,z)$ for the given derivatives.

In order to find $f(x,y,y)$ we integrate with respect to x the first of the equations

$\frac{\partial}{\partial x} f(x,y,z) = yz$ (eq1)

$\frac{\partial}{\partial y} f(x,y,z) = xz + z$ (eq2)

$\frac{\partial}{\partial z} f(x,y,z) = xy + y + 3z^2$ (eq3)

and get

$f(x,yz) = \int yzdx = xyz + C_1(y,z)$

Thus

$f(x,y,z) = xyz + C_1(y,z)$ ( i )

We have integrated with respect to *x*, therefore the constant still depend on *y* and *z*.

Now, to find $C_1(y,z)$ we differentiate and substitute (i) in (eq2) and get:

$$\frac{\partial}{\partial y}\left(xyz + C_1(y,z)\right) = xz + z \Rightarrow xz + \frac{\partial}{\partial y}\left(C_1(y,z)\right) = xz + z \Rightarrow$$

$$\frac{\partial}{\partial y}\left(C_1(y,z)\right) = z \Rightarrow C_1(y,z) = yz + C_2(z).$$

(We have integrated with respect to *y* , therefore the constant still depend on and *z*.

Thus $f(x,y,z) = xyz + yz + C_2(z)$ (ii)

Now , substituting (ii) in (eq3) we have

$$\frac{\partial}{\partial z}(xyz + yz + C_2(z)) = xy + y + 3z^2 \Rightarrow$$

$$xy + y + \frac{\partial}{\partial z}(C_2(z)) = xy + y + 3z^2 \Rightarrow$$

$$\frac{\partial}{\partial z}(C_2(z)) = 3z^2$$

$$C_2(z) = z^3 + C$$

Finally, substituting $C_2(z) = z^3 + C$ in (ii) we have

$f(x,y,z) = xyz + yz + z^3 + C$ (where C is a constant).

**13.**

**Answer:**

From

$$\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x} f\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y} f\right) \Rightarrow ax^2 z = 3x^2 z \Rightarrow a = 3$$

$$\frac{\partial}{\partial z}\left(\frac{\partial}{\partial x} f\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial z} f\right) \Rightarrow x^3 = bx^3 \Rightarrow b = 1$$

$$\frac{\partial}{\partial z}\left(\frac{\partial}{\partial y} f\right) = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial z} f\right) \Rightarrow x^3 = bx^3 \Rightarrow b = 1.$$

Thus, all three conditions are fulfilled if $a = 3$ and $b = 1$.

For these values of *a* and *b* we get $f(x,y,z) = x^3 yz + x^2 y + z^2 + C$

**Calculation of the pressure field for a known velocity field for an incompressible, steady state, isothermal Newtonian flow.**

**14.**

## Answer:

**a)** $P = -\rho g z - 8\rho x^2 - 8\rho y^2 + C$

**b)** $P = -\rho g z - \frac{17}{2}\rho x^2 - \frac{17}{2}\rho y^2 + C$

**c)** $P = -\rho g z + 4\rho x^2 + 4\rho y^2 + 4\rho y + C$

**Solution a)**

We substitute $u = 2x + 3y, \quad v = 4x - 2y, \quad w = 0$ in eq1,2,3,4 and get ( *note that al derivatives with respect to t are 0*):

**Continuity equation:**

$0 = 0$ **eq1i.** **(** *identically fulfilled***)**

**Navier Stokes equations:**

*x component:*

$16\rho x = -\frac{\partial P}{\partial x}$ **eq2i.**

*y component:*

$16\rho y = -\frac{\partial P}{\partial y}$ **eq3i.**

*z component:*

$0 = -\frac{\partial P}{\partial z} - \rho g$ **eq4i.**

Now **eq2i.** gives $P(x, y, z) = -8\rho x^2 + C_1(y, z)$ (*) .

Substitution in **eq3i.** implies

$16\rho y = -\frac{\partial C_1(y, z)}{\partial y} \Rightarrow$

$C_1(y, z) = -8\rho y^2 + C_2(z)$

Hence, from (*) we have

$P(x, y, z) = -8\rho x^2 - 8\rho y^2 + C_2(z)$ (**)

Now we substitute (**) in **eq4i.** and get

$0 = -\frac{\partial P}{\partial z} - \rho g \Rightarrow 0 = -\frac{\partial}{\partial z}(C_2(z)) - \rho g$

$\Rightarrow C_2(z) = -\rho g z + C$ ( where C is a constant)

Finally, substituting $C_2(z) = -\rho g z + C$ in (**) we have

$P(x, y, z) = -8\rho x^2 - 8\rho y^2 - \rho g z + C$ (where C is a constant).

**15.**

## Solution A:

$\frac{\partial(\rho\varphi)}{\partial t} + \nabla \bullet (\rho\varphi\vec{U}) = \nabla \bullet (\Gamma \cdot (\nabla\varphi)) + \nabla \bullet (\nabla \times \vec{U}) + 16x + 16y + 4xz + 8yz - 24 \Rightarrow$

$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi\vec{U}) = div(\Gamma grad\varphi) + div(curl(\vec{U})) + 16x + 16y + 4xz + 8yz - 24 \Rightarrow$

(since $curl(\vec{U}) = (2, -1, 0)$ we have $div(curl(\vec{U})) = 0$ )

$\frac{\partial(\rho\varphi)}{\partial t} + div(\rho\varphi u, \rho\varphi v, \rho\varphi w) = div(\Gamma\frac{\partial\varphi}{\partial x}, \Gamma\frac{\partial\varphi}{\partial y}, \Gamma\frac{\partial\varphi}{\partial z}) + 16x + 16y + 4xz + 8yz - 24 \Rightarrow$

$$\frac{\partial(\rho\varphi)}{\partial t}+\frac{\partial(\rho\varphi u)}{\partial x}+\frac{\partial(\rho\varphi v)}{\partial y}+\frac{\partial(\rho\varphi w)}{\partial z})=\frac{\partial}{\partial x}\left(\Gamma\frac{\partial\varphi}{\partial x}\right)+\frac{\partial}{\partial y}\left(\Gamma\frac{\partial\varphi}{\partial y}\right)+\frac{\partial}{\partial z}\left(\Gamma\frac{\partial\varphi}{\partial z}\right)+16x+16y+4xz+8yz-24$$

(eq2)

We substitute $\rho=2$, , $\vec{U}=(4,4,x+2y)$ and $\varphi(x,y,z)=1+3t+x^2+y^2+z^2$ in the equation (eq2) and get

$$\frac{\partial(2\varphi)}{\partial t}+\frac{\partial(8\varphi)}{\partial x}+\frac{\partial(8\varphi)}{\partial y}+\frac{\partial(2(x+2y)\phi)}{\partial z})=\frac{\partial}{\partial x}\left(\Gamma\frac{\partial\varphi}{\partial x}\right)+\frac{\partial}{\partial y}\left(\Gamma\frac{\partial\varphi}{\partial y}\right)+\frac{\partial}{\partial z}\left(\Gamma\frac{\partial\varphi}{\partial z}\right)+16x+16y+4xz+8yz-24$$

( Note that $\Gamma$ is a constant)

$6+16x+16y+4xz+8yz=6\Gamma+16x+16y+4xz+8yz-24\Rightarrow$

$30=6\Gamma\Rightarrow$

$\Gamma=5$

**Answer A:** $\Gamma=5$

## Solution B:

We substitute $u=6+4x,\quad v=4-2y,\quad w=2-2z$ in eq1,2,3,4 and get ( *note that al derivatives with respect to t are 0*):

**Continuity equation:**

$0=0$ **eq1i.** **(** *identically fulfilled***)**

**Navier Stokes equations:**

*x component:*

$$\rho(16x+24)=-\frac{\partial P}{\partial x}$$ **eq2i.**

*y component:*

$$\rho(4y-8)=-\frac{\partial P}{\partial y}$$ **eq3i.**

*z component:*

$$\rho(4z-4)=-\frac{\partial P}{\partial z}-\rho g$$ **eq4i.**

Now **eq2i.** gives $P(x,y,z)=\rho(-8x^2-24x)+C_1(y,z)$ (*) .

Substitution in **eq3i.** implies

$$\rho(4y-8)=-\frac{\partial C_1(y,z)}{\partial y}\Rightarrow$$

$C_1(y,z)=\rho(-2y^2+8y)+C_2(z)$

Hence, from (*) we have

$P(x,y,z)=\rho(-8x^2-24x)+\rho(-2y^2+8y)+C_2(z)$ (**)

Now we substitute (**) in **eq4i.** and get

$$\rho(4z-4)=-\frac{\partial P}{\partial z}-\rho g\Rightarrow\rho(4z-4)=-\frac{\partial}{\partial z}(C_2(z))-\rho g$$

$\Rightarrow C_2(z)=-\rho gz+\rho(-2z^2+4z)+C$ ( where C is a constant)

Finally, substituting $C_2(z)=-\rho gz+C$ in (**) we have

$P(x,y,z)=\rho(-8x^2-24x)+\rho(-2y^2+8y)+\rho(-2z^2+4z)-\rho gz+C$

**Answer B:**

$P(x,y,z)=\rho(-8x^2-24x-2y^2+8y-2z^2+4z-gz)+C$
(where C is a constant).

## 16. Solution

$\vec{V}=(2+3x,5-y,1-az)$
First we substitute $u=2+3x,\quad v=5-y,\quad w=1-az$ in eq1
and get ( *note that al derivatives with respect to t are 0*):

**Continuity equation:**
$3-1-a=0\Rightarrow a=2$
No we have $\vec{V}=(2+3x,5-y,1-2z)$
**Using the Navier Stokes equations we get:**
*x component:*

$$\rho(9x+6)=-\frac{\partial P}{\partial x}$$ **eq2i.**

*y component:*

$$\rho(y-5)=-\frac{\partial P}{\partial y}$$ **eq3i.**

*z component:*

$$\rho(4z-2)=-\frac{\partial P}{\partial z}-\rho g$$ **eq4i.**

Now **eq2i.** gives $P(x,y,z)=\rho(\frac{-9x^2}{2}-6x)+C_1(y,z)$ (*) .
Substitution in **eq3i.** implies

$$\rho(y-5)=-\frac{\partial C_1(y,z)}{\partial y}\Rightarrow$$

$$C_1(y,z)=\rho(\frac{-y^2}{2}+5y)+C_2(z)$$

Hence, from (*) we have

$$P(x,y,z)=\rho(\frac{-9x^2}{2}-6x)+\rho(\frac{-y^2}{2}+5y)+C_2(z)\qquad (**)$$

We substitute (**) in **eq4i.** and get

$$\rho(4z-2)=-\frac{\partial P}{\partial z}-\rho g\Rightarrow\rho(4z-2)=-\frac{\partial}{\partial z}(C_2(z))-\rho g$$

$\Rightarrow C_2(z)=-\rho gz+\rho(-2z^2+2z)+C$ ( where C is a constant)
Finally, substituting $C_2(z)$ in (**) we have

$$P(x,y,z)=\rho(\frac{-9x^2}{2}+6x)+\rho(\frac{-y^2}{2}+5y)+-\rho gz+\rho(-2z^2+2z)+C$$

**Answer :**

$$P(x,y,z)=\rho(-\frac{9x^2}{2}+6x-\frac{y^2}{2}+5y-gz-2z^2+2z)+C$$

(where C is a constant).

**Q17.** Consider steady, incompressible, isothermal, laminar stationary Newtonian flow in a long round pipe in the z-direction, with constant circular cross-section of radius R=2 m. Use

the continuity and the Navier-Stokes equations in cylindrical coordinates to find the velocity field $\mathbf{V}=(u_r, u_\theta, u_z)$ and the pressure field $P(r,\theta,z)$ if the fluid flow satisfies the following conditions:

**c0.** All partial derivatives with respect to time t are 0 **( Steady flow)**

**c1.** $\mu=0.001$ kg/(m·s) and $\rho=1000$ kg/m$^3$

**c2.** A Constant pressure gradient $\partial P/\partial z = -1/250$ Pa/m is applied in the horizontal axis ( z-axis in our notation):
$\partial P/\partial z = -1/250$,

**c3.** The flow is parallel to the z axis, that is $u_r=0$ and $u_\theta=0$.

**c4.** We assume that the flow is axisymmetric . The velocity does not depend on θ, that is $\frac{\partial u_z}{\partial \theta}=0$

**c5.** Boundary cond. 1 ( No-slip boundary condition, $\mathbf{V}_{fluid}=\mathbf{V}_{wall}$ ): If r=2 then $u_z=0$

**c6.** Boundary condition 2: $u_z$ has maximum at r=0 that is $\left.\frac{\partial u_z}{\partial r}\right|_{r=0}=0$

*The continuity and the Navier-Stokes equations for an incompressible , isothermal Newtonian flow (density $\rho$ =const, viscosity $\mu$ =const), with a velocity field* $\vec{V}=(u_r,u_\theta,u_z)$ ***in Cylindrical coordinates*** $(r,\theta,z)$*:*

**----------**

SOLUTION

**Incompressible continuity equation**

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r}+\frac{1}{r}\frac{\partial(u_\theta)}{\partial \theta}+\frac{\partial u_z}{\partial z}=0 \qquad \textbf{eq a)}$$

**Navier-Stokes equations in Cylindrical coordinates:**

*r-component:*

$$\rho\left(\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta}-\frac{u_\theta^2}{r}+u_z\frac{\partial u_r}{\partial z}\right)$$

$$=-\frac{\partial P}{\partial r}+\rho g_r+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right)-\frac{u_r}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2}-\frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}+\frac{\partial^2 u_r}{\partial z^2}\right] \qquad \textbf{eq b)}$$

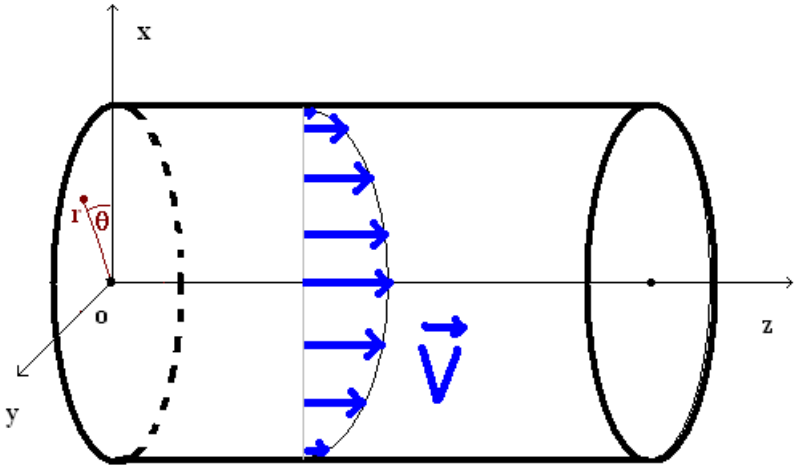
*$\theta$-component:*

$$\rho\left(\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r u_\theta}{r}+u_z\frac{\partial u_\theta}{\partial z}\right)$$

$$=-\frac{1}{r}\frac{\partial P}{\partial \theta}+\rho g_\theta+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\theta}{\partial r}\right)-\frac{u_\theta}{r^2}+\frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2}+\frac{2}{r^2}\frac{\partial u_r}{\partial \theta}+\frac{\partial^2 u_\theta}{\partial z^2}\right] \qquad \textbf{eq c)}$$

*z-component:*

$$\rho\left(\frac{\partial u_z}{\partial t}+u_r\frac{\partial u_z}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta}+u_z\frac{\partial u_z}{\partial z}\right)$$
$$=-\frac{\partial P}{\partial z}+\rho g_z+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2}+\frac{\partial^2 u_z}{\partial z^2}\right] \qquad \textbf{eq d)}$$




We choose x as a vertical axis, y an z are in a horizontal plane and the flow is parallel with the z-axis. We denote velocity vector **V**=($u_r$, $u_\theta$, $u_z$) where $u_r$, $u_\theta$ and $u_z$ are r-component, θ-component and z-component in cylindrical coordinates. According to the assumptions we have $u_r$ =0, $u_\theta$ = 0, and $u_z$ does not depend on θ.
Since x is the vertical axis we have that vector g=(-g, 0,0) where g=9,81 m/s$^2$ which in cylindrical coordinates gives

$g_r=-g\cos\theta$, $g_\theta=g\sin\theta$ and $g_z=0$

Now we substitute $\partial P/\partial z = -1/250$ Pa/m, μ=0.001kg /(ms) in the continuity and Navier-Stokes equations:

Since $u_r$ =0 and $u_\theta$ =0 (according to c3), continuity equation in cylindrical coordinates

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r}+\frac{1}{r}\frac{\partial(u_\theta)}{\partial \theta}+\frac{\partial u_z}{\partial z}=0$$

gives

$$\frac{\partial u_z}{\partial z}=0\,.$$

This tells us that $u_z$ is not a function of z. Furthermore, since $u_z$ velocity does not depend on θ (assumption c4) we conclude that $u_z$ depends only on r.
To simplify notation we denote

$u_z = w(r)$ (*)

Now we substitute

$g_r = -g\cos\theta$, $g_\theta = g\sin\theta$ and $g_z = 0$

$\partial P/\partial z = -1/250$ Pa/m, $\mu$=0.001kg /(ms)

in the Navier-Stokes equations:

The r-component of the Navier-Stokes equation gives:

$$0 = -\frac{\partial P}{\partial r} - \rho g\cos\theta \qquad \textbf{( eq r-c)}$$

The θ-component of the Navier-Stokes equation:

$$0 = -\frac{1}{r}\frac{\partial P}{\partial \theta} + \rho g\sin\theta \qquad \textbf{( eq θ-c)}$$

The Z-component of the Navier-Stokes equation (where $u_z = w(r)$ and $\frac{\partial P}{\partial z} = -\frac{1}{250}$) givs:

$$0 = \frac{1}{250} + \frac{1}{1000}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right)\right] \qquad \textbf{( eq z-c)}$$

**Step 1.** We find the pressure $P = P(r,\theta,z)$ .

In order to find the pressure $P$ we solve **( eq r-c), ( eq θ-c)** and the equation $\frac{\partial P}{\partial z} = -\frac{1}{250}$ that is

$$\frac{\partial P}{\partial r} = -\rho g\cos\theta$$

$$\frac{\partial P}{\partial \theta} = +r\rho g\sin\theta$$

$$\frac{\partial P}{\partial z} = -\frac{1}{250}$$

From these equations we get

$$P = -\frac{1}{250}z - \rho g r\cos\theta + C$$

**Step 2**. We find the velocity component $u_z = w(r)$.

We solve **( eq z-c)** with boundaries c5 and c6:

$$0 = \frac{1}{250} + \frac{1}{1000}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right)\right] \qquad \textbf{( eq z-c)}$$

$$w(2) = 0 \qquad \textbf{(c5)}$$

$$\left.\frac{\partial w}{\partial r}\right|_{r=0} = 0 \qquad \textbf{(c6)}$$

( **Remark:** Technically, we can write $\frac{dw}{dr}$ instead $\frac{\partial w}{\partial r}$ since *w* is now a function of only one variable)

From **( eq z-c)** we have

$$0 = \frac{1}{250} + \frac{1}{1000}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right)\right]$$

$$\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right) = -4r \Rightarrow$$

$$r\frac{\partial w}{\partial r} = -2r^2 + C_1 \Rightarrow \text{ (substitution } r = 0 \text{ and (c6)} \Rightarrow C_1 = 0\ )$$

$$r\frac{\partial w}{\partial r} = -2r^2 \Rightarrow$$

$$\frac{\partial w}{\partial r} = -2r \Rightarrow$$

$$w = -r^2 + C_2 \Rightarrow \text{ (substitution } r = 2 \text{ and (c5)} \Rightarrow C_2 = 4\ )$$

$$w = -r^2 + 4 \Rightarrow$$

Thus $u_z = w(r) = -r^2 + 4$ and

$\mathbf{V} = (u_r, u_\theta, u_z) = (0,\ 0,\ -r^2 + 4)$.

**Answer :**

$$P = -\frac{1}{250}z - \rho g r\cos\theta + C\ ,$$

$$\mathbf{V} = (0,\ 0,\ -r^2 + 4)$$

**18.**

We consider an incompressible ( density $\rho$ =const), steady state ( variables do not depend on time), isothermal Newtonian flow with a given velocity field

$\vec{V} = (u(x,y,z),\ v(x,y,z),\ w\ (x,y,z)) = (3x + cy,\ 4 + 2x - y + bz,\ 3 - az)$

Use the following equations ( continuity and Navier Stokes equations) , where

$\rho$ =constant, $\mu$ =constant , $\vec{g} = (0,0,-g)$ i.e. $g_x = g_y$ =0 and $g_z = -g$ ( $g \approx 9.81 m/s^2$)

to find:

i) parameters a, b and c

ii) en expression for pressure P(*x,y,z*) as a function of *x,y* and *z*.

**Incompressible continuity equation:**

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \qquad \textbf{eq1.}$$

**Navier Stokes equations:**

*x component:*

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right) = -\frac{\partial P}{\partial x} + \rho g_x + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) \qquad \textbf{eq2.}$$

*y component:*

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right) = -\frac{\partial P}{\partial y} + \rho g_y + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) \qquad \textbf{eq3.}$$

*z component:*

$$\rho\left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) \qquad \textbf{eq4.}$$

---

We substitute $u = 3x + cy,\quad v = 4 + 2x - y + bz,\quad w = 3 - az,$ in eq1,2,3,4 and get ( *note that al derivatives with respect to t are 0*):

**Continuity equation:**

$2 - a = 0 \Rightarrow a = 2$ **eq1i.**

Thus

$u = 3x + cy,\quad v = 4 + 2x - y + bz,\quad w = 3 - 2z,$

**Navier Stokes equations:**

*x component:*

$$9\,\rho\,x + 2\,\rho\,c\,y + 4\,\rho\,c + 2\,\rho\,c\,x + \rho\,c\,b\,z = -\left(\frac{\partial}{\partial x}\,P(x,y,z)\right) \qquad \textbf{eq2i.}$$

*y component:*

$$4\,\rho\,x + 5\,\rho\,y - 4\,\rho = -\left(\frac{\partial}{\partial y}\,P(x,y,z)\right) \qquad \textbf{eq3i.}$$

*z component:*

$-6\,\rho + 4\,\rho\, z = -\rho\, g - \left(\frac{\partial}{\partial z} P(x,y,z)\right)$ **eq4i.**

The system (eq2i, eq3i, eq4i) is solvable only if mixed derivatives are equal:

$$Con1: \quad \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}P\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}P\right) \Rightarrow -2\rho c = -4\rho \Rightarrow c = 2$$

$$Con2: \quad \frac{\partial}{\partial z}\left(\frac{\partial}{\partial x}P\right) = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial z}P\right) \Rightarrow \rho bc = 0 \Rightarrow b = 0$$

$$Con3: \quad \frac{\partial}{\partial z}\left(\frac{\partial}{\partial y}P\right) = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial z}P\right) \Rightarrow -3\rho b = 0 \Rightarrow b = 0$$

Thus c=2 and b=0
We solve simplified equations

$13\,\rho\, x + 4\,\rho\, y + 8\,\rho = -\left(\frac{\partial}{\partial x} P(x,y,z)\right)$

$4\,\rho\, x + 5\,\rho\, y - 4\,\rho = -\left(\frac{\partial}{\partial y} P(x,y,z)\right)$

$-6\,\rho + 4\,\rho\, z = -\rho\, g - \left(\frac{\partial}{\partial z} P(x,y,z)\right)$

and get

$$P(x,y,z) = \rho(-\frac{13x^2}{2} - \frac{5y^2}{2} - 2z^2 - 4xy - 8x + 4y - gz + 6z) + C$$

**Answer.** $P(x,y,z) = \rho(-\frac{13x^2}{2} - \frac{5y^2}{2} - 2z^2 - 4xy - 8x + 4y - gz + 6z) + C$

**19.** We consider a scalar field $f(r,\theta,z)$ given in cylindrical coordinates, where $x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$, and basis vectors are $\vec{e}_1, \vec{e}_2, \vec{e}_3$.
Find the expression for the gradient, $grad(f(r,\theta,z))$, in cylindrical coordinates, that is in terms of $r, \theta, z, \vec{e}_1, \vec{e}_2, \vec{e}_3, \frac{\partial f}{\partial r}, \frac{\partial f}{\partial \theta}$ and $\frac{\partial f}{\partial z}$ if

**a)** $\vec{e}_1 = \vec{i}, \quad \vec{e}_2 = \vec{j}, \quad \vec{e}_3 = \vec{k}$ ( we keep the same basis $\vec{i}, \vec{j}, \vec{k}$ ) .

**b)** $\vec{e}_1 = \vec{i} \quad \vec{e}_2 = 2\vec{j}, \quad \vec{e}_3 = 2\vec{j} + \vec{k}$

**c)** $\vec{e}_1 = \vec{i}\cos\theta, \quad \vec{e}_2 = \vec{j}\sin\theta, \ \vec{e}_3 = \vec{k}$ **(exam 1, 2012; Q5 B (2 points))**

**d)** $\vec{e}_1 = \cos\theta\,\vec{i} + \sin\theta\,\vec{j}, \quad \vec{e}_2 = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}, \quad \vec{e}_3 = \vec{k}$ (this is often used as a local basis for cylindrical coordinates)

## Solution:

In x,y, z variables we have

$$grad(f)=(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y},\frac{\partial f}{\partial z})=\frac{\partial f}{\partial x}\vec{i}+\frac{\partial f}{\partial y}\vec{j}+\frac{\partial f}{\partial z}\vec{k} \qquad (\text{eq1})$$

For cylindrical coordinates we have $x=r\cos\theta$, $y=r\sin\theta$, z=z

First we write the derivatives $\frac{\partial f}{\partial x}$, and $\frac{\partial f}{\partial y}$ in $r,\theta$ coordinates (the variable z is in both coord systems) .

Solving the following system for $\frac{\partial f}{\partial x}$, and $\frac{\partial f}{\partial y}$,

$$\frac{\partial f}{\partial r}=\frac{\partial f}{\partial x}\cdot\cos\theta+\frac{\partial f}{\partial y}\cdot\sin\theta$$

$$\frac{\partial f}{\partial \theta}=\frac{\partial f}{\partial x}\cdot(-r\sin\theta)+\frac{\partial f}{\partial y}\cdot(r\cos\theta)$$

we get

$$\frac{\partial f}{\partial x}=\cos\theta\frac{\partial f}{\partial r}-\frac{\sin\theta}{r}\frac{\partial f}{\partial \theta}$$

$$\frac{\partial f}{\partial y}=\sin\theta\frac{\partial f}{\partial r}+\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta} \qquad (**)$$

We substitute the derivatives (**) in ( eq1) and get

$$grad(f)=(\cos\theta\frac{\partial f}{\partial r}-\frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{i}+(\sin\theta\frac{\partial f}{\partial r}+\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\vec{j}+\frac{\partial f}{\partial z}\vec{k} \qquad (***)$$

To solve problems a) , b) c) and d) we must express $\vec{i},\vec{j},\vec{k}$ as a linear combinations of $\vec{e}_1,\vec{e}_2,\vec{e}_3$ and substitute them into (***) .

a)

From (***), since $\vec{i}=\vec{e}_1$, $\vec{j}=\vec{e}_2$, $\vec{k}=\vec{e}_3$, we have immediately

$$grad(f)=(\cos\theta\frac{\partial f}{\partial r}-\frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_1+(\sin\theta\frac{\partial f}{\partial r}+\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_2+\frac{\partial f}{\partial z}\vec{e}_3$$

b) From $\vec{e}_1=\vec{i}$ $\vec{e}_2=2\vec{j}$, $\vec{e}_3=2\vec{j}+\vec{k}$ we find

$$\vec{i}=\vec{e}_1,\quad \vec{j}=\frac{\vec{e}_2}{2},\quad \vec{k}=\vec{e}_3-\vec{e}_2 \qquad (\text{eq b})$$

Then we put $\vec{i},\vec{j},\vec{k}$ from ( eq b) into (***) and get

$$grad(f)=(\cos\theta\frac{\partial f}{\partial r}-\frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_1+(\sin\theta\frac{\partial f}{\partial r}+\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\frac{\vec{e}_2}{2}+\frac{\partial f}{\partial z}(\vec{e}_3-\vec{e}_2)$$

and, after collecting components for $\vec{e}_1 \quad \vec{e}_2, \quad \vec{e}_3$

$$grad(f) = (\cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_1 + (\frac{1}{2}\sin\theta \frac{\partial f}{\partial r} + \frac{1}{2}\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta} - \frac{\partial f}{\partial z})\vec{e}_2 + \frac{\partial f}{\partial z}\vec{e}_3$$

**c)** From

$$\vec{e}_1 = \vec{i}\cos\theta, \quad \vec{e}_2 = \vec{j}\sin\theta\,,\ \vec{e}_3 = \vec{k}$$

we have

$$\vec{i} = \frac{\vec{e}_1}{\cos\theta}, \quad \vec{j} = \frac{\vec{e}_2}{\sin\theta}, \quad \vec{k} = \vec{e}_3 \ \ (\text{eq c})$$

Putting $\vec{i}, \vec{j}, \vec{k}$ from (eq c) into (***) gives

$$grad(f) = (\cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\frac{\vec{e}_1}{\cos\theta} + (\sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\frac{\vec{e}_2}{\sin\theta} + \frac{\partial f}{\partial z}\vec{e}_3$$

**d)** $\vec{e}_1 = \cos\theta\,\vec{i} + \sin\theta\,\vec{j}, \quad \vec{e}_2 = -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}, \quad \vec{e}_3 = \vec{k}$.

We can solve d) in the same manner as in a,b,c but this time we can just collect terms $\frac{\partial f}{\partial r}$, $\frac{\partial f}{\partial \theta}$, $\frac{\partial f}{\partial z}$ and get the result:

$$grad(f) = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k} = (\cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{i} + (\sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\vec{j} + \frac{\partial f}{\partial z}\vec{k}$$

$$= \frac{\partial f}{\partial r}\cdot(\cos\theta\,\vec{i} + \sin\theta\,\vec{j}) + \frac{1}{r}\frac{\partial f}{\partial \theta}\cdot(-\sin\theta\,\vec{i} + \cos\theta\,\vec{j}) + \frac{\partial f}{\partial z}\vec{k}$$

$$= \frac{\partial f}{\partial r}\cdot\vec{e}_1 + \frac{1}{r}\frac{\partial f}{\partial \theta}\cdot\vec{e}_2 + \frac{\partial f}{\partial z}\vec{e}_3$$

**Answer:**

**a)** $grad(f) = (\cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_1 + (\sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_2 + \frac{\partial f}{\partial z}\vec{e}_3$

**b)** $grad(f) = (\cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\vec{e}_1 + (\frac{1}{2}\sin\theta \frac{\partial f}{\partial r} + \frac{1}{2}\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta} - \frac{\partial f}{\partial z})\vec{e}_2 + \frac{\partial f}{\partial z}\vec{e}_3$

**c)** $grad(f) = (\cos\theta \frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta})\frac{\vec{e}_1}{\cos\theta} + (\sin\theta \frac{\partial f}{\partial r} + \frac{\cos\theta}{r}\frac{\partial f}{\partial \theta})\frac{\vec{e}_2}{\sin\theta} + \frac{\partial f}{\partial z}\vec{e}_3$

**d)** $grad(f(r,\theta,z) = \frac{\partial f}{\partial r}\cdot\vec{e}_1 + \frac{1}{r}\frac{\partial f}{\partial \theta}\cdot\vec{e}_2 + \frac{\partial f}{\partial z}\vec{e}_3$

**20.** See
http://ingforum.haninge.kth.se/armin/AR_2000/HL2008/DERIV_NAVIER_STOKES.pdf

**21.** Solution:

$$z = 1 + 3x + 2y$$

$$\vec{N} = (-z'_x, -z'_y, 1) = (-3, -2, 1)$$

$$\vec{F} = (x, y, x + y)$$

$$\vec{F} \circ \vec{N} = -3x - 2y + x + y = -2x - y$$

$$\Phi = \iint_D \vec{F} \circ \vec{N} dxdy = \int_0^1 dx \int_0^2 (-2x - y)dy = \int_0^1 (-4x - 2)dx = -4$$

Answer: $\Phi = -4$

**22.** Solution:

$$div(\vec{F}) = 12 + 2z - 2z = 12$$

$$\Phi = \iiint_K div\vec{F}dV = \iiint_K 12dxdydz = 12 \times \text{Volume(K)} = 12 \times \frac{4}{3} \times 3^3 \pi = 432\pi$$

Answer: $\Phi = 432\pi$